

Python Stock Analysis is a lightweight, modular toolkit designed to support fundamental equity analysis using Python. The project focuses on transparency, composability, and ease of extension, providing a clean foundation for investors or analysts who want to evaluate stocks based on financial fundamentals rather than technical indicators. The library integrates market price data and company financial statements through the yfinance API and exposes reusable analytics, screening, simulation, and visualization utilities suitable for scripts or Jupyter notebooks.

The package is organized into clearly separated modules that reflect the analysis workflow. Core models define structured containers such as Stock metadata, FundamentalsSnapshot summaries, valuation results, and portfolio-level statistics, along with enums that standardize scoring styles, valuation methods, rebalance frequencies, and return intervals. The data layer is responsible for fetching historical prices, current prices, benchmark indices, and company financial statements including income statements, cash flow statements, and balance sheets. All data access functions are defensive by design: network or API failures are logged and return empty data structures, allowing downstream analysis to fail gracefully.

The analysis layer provides the core fundamental logic. It includes utility helpers for safe arithmetic, ratio computation, compound annual growth rates, and margin calculations. From these building blocks, the toolkit derives higher-level metrics such as P/E ratios, P/FCF ratios, free cash flow yield, revenue and cash flow growth, and profitability margins. These metrics are combined into intuitive value, growth, and quality scores, which are then aggregated into a composite score. An end-to-end `summarize_fundamentals` workflow pulls raw financial data and produces a single FundamentalsSnapshot object that captures the company's valuation, financial health, growth profile, and scoring as of a specific date.

For valuation analysis, the project supports time-series P/E analysis by aligning historical prices with a dynamically constructed EPS history derived from quarterly and annual statements, with a trailing EPS fallback when necessary. This enables the creation of historical P/E series, rolling P/E

averages for smoothing, and descriptive statistics such as mean, median, and extrema. These tools support relative valuation analysis over time rather than relying solely on point-in-time multiples.

The screener module enables quick fundamental filtering across multiple tickers by applying constraints such as minimum free cash flow yield, maximum P/E, and leverage limits. Screen results are returned as tidy DataFrames that can be ranked, compared, or visualized. In addition, the investment module provides long-term planning tools, including deterministic dollar-cost averaging growth paths and Monte Carlo simulations of monthly contributions using stochastic return assumptions. Summary statistics from these simulations quantify expected outcomes and downside or upside risk.

Visualization helpers are built on Matplotlib and return figure objects for easy saving or further customization. Supported charts include P/E series with rolling averages, screener comparison bar charts, DCA balance growth curves, and Monte Carlo outcome distributions. Supporting utilities provide structured logging and configuration management via JSON files.

Overall, the Python Stock Analysis project delivers a cohesive, extensible framework for fundamental stock analysis, screening, valuation context, and long-term investment simulation. Its modular design, defensive data handling, and emphasis on clarity make it well suited as both a practical analysis toolkit and a foundation for more advanced fundamental research.